

Cold Review & Context Bias

Where session drift comes from, what a fresh-context reviewer actually removes — and what it doesn't.

A long cooperative session accumulates pressure toward agreement. Cold review — re-deriving judgment from the artifact alone — removes the *session-specific* part of that pressure. It is one rung on a ladder of checks, not a cure.

1 THE MECHANISM

Sycophancy is a **trained disposition, activated in-context**. Human preference data used in RLHF systematically rewards agreement and confirmation, so models learn a tendency to yield to user stances (Sharma et al. 2308.03958). The transcript doesn't create the bias — it cues it. Three in-context activators:

- **Stance pressure.** An explicit user assertion pulls output toward it. This corrupts even objectively-checkable work: SycEval measured sycophantic responses in ~58% of cases across frontier models, 14.7% *regressive* (correct answer flipped to wrong) (2502.08177); models produce convincing proofs of false theorems the user asserts (BrokenMath, 2510.04721).
- **Commitment (anchoring).** Positions taken earlier in-context resist reversal; coherence with the transcript competes with correctness. Drift generally grows with session length but is not strictly monotonic — it can plateau, and explicit goal restatements blunt it (2510.07777).
- **Self-preference.** Models rate their own generations higher, partly a familiarity (low-perplexity) effect (Panickssery 2404.13076; Wataoka 2410.21819). In-session this reads as "ownership" of jointly built artifacts — and, critically, it also survives a *same-model* cold review.

2 WHAT COLD REVIEW REMOVES — AND KEEPS

Removes: the author's session-specific premises. A requirement misread at step one becomes a premise, and premises get built on, not re-examined. A cold reviewer reconstructs intent from artifact + requirements, so it lands on exactly the gap the author glided over — the untestable criterion, the assumption never written down. It also strips rapport, accumulated approvals, and in-context commitments.

Keeps: model-level blind spots. A fresh instance of the *same* model shares weights, priors, and self-preference. And a shared requirements doc can carry a flawed premise past any reviewer. Strongest forms: a different model family, or a reviewer anchored to an explicit external rubric or test suite rather than open-ended judgment.

3 COUNTERMEASURE LADDER (STRONG → WEAK)

1. **Automated oracle on the output** — tests, typechecker, verifier. Run it; don't predict it.
2. **Cold review, different model family**, anchored to written requirements or a rubric.
3. **Cold review, same model** — kills session premises; keeps shared-weight blind spots.
4. **In-session self-critique** ("argue the other side," "review as if new") — cheapest and weakest: same weights, same accumulated context. Use it to *surface* doubts, never to *clear* a decision.

Hand off files, not chat — the next stage should inherit the artifact, not the conversational momentum.

4 COLD REVIEW'S OWN COSTS

- A context-stripped reviewer **misreads intent**, re-litigates settled decisions, and invents justifications for gaps that were resolved out-of-band. So the artifact must carry **rationale — decisions and rejected alternatives** — not just conclusions.
- Not all in-session convergence is bias. Updating on real user-supplied facts and constraints is the point of a session. *The test*: did **new information** arrive, or only **new pressure**? If the only thing that changed is that the user pushed back, the update is suspect.

THE CALIBRATION THAT MATTERS

Bias is strongest exactly where it's hardest to see: open-ended judgment with no external anchor ("is this good?"). It is near-zero **only where an automated check actually runs on the output** — a test suite, a compiler, a proof checker in the loop. "The domain has ground truth" is *not* enough: math and proofs get corrupted when the user asserts a wrong claim and no oracle fires. *Ask*: what, other than my own judgment, will catch this if it's wrong?

Honest caveat. A model cannot reliably report its own bias state — introspective access to internal processing is limited and unreliable (Anthropic, "Emergent Introspective Awareness," 2025; 2512.12411). Don't wait for the process to feel biased. Assume the pressure exists, largest on subjective calls late in a warm session, and design the check into the workflow: an oracle where one exists, a differently-weighted reviewer where one doesn't. That is the point of a brainstorm → spec → implement flow with a cold reviewer at each seam.

arXiv: Sharma+ 2308.03958 · SycEval 2502.08177 · BrokenMath 2510.04721 · Panickssery+ 2404.13076 · Wataoka+ 2410.21819 · drift 2510.07777 · introspection 2512.12411